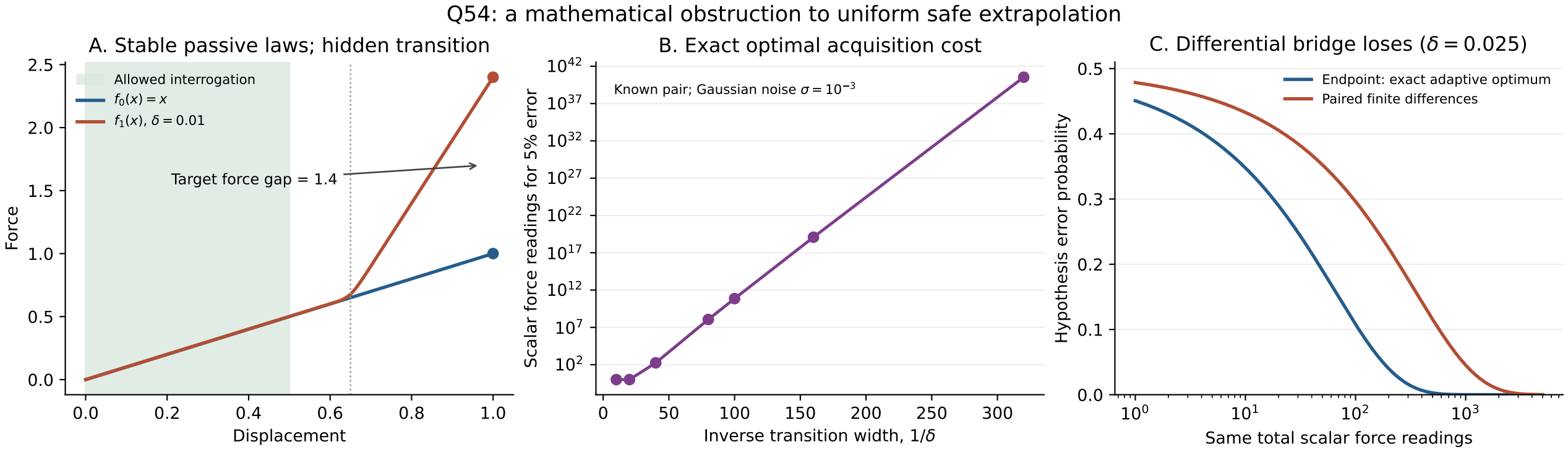
Q54: a mathematical obstruction to uniform safe extrapolation
A. Stable passive laws; hidden transition
Allowed interrogation
$f_0(x) = x$
$f_1(x)$, $\delta = 0.01$
Target force gap = 1.4
Force
Displacement
B. Exact optimal acquisition cost
Known pair; Gaussian noise $\sigma = 10^{-3}$
Scalar force readings for 5% error
Inverse transition width, $1/\delta$
C. Differential bridge loses ($\delta = 0.025$)
Endpoint: exact adaptive optimum
Paired finite differences
Hypothesis error probability
Same total scalar force readings